

Replication Report

Clustering Huge Protein Sequence Sets in Linear Time
OSTI 1624105 (Steinegger & Söding, *Nature Communications* 2018)

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Abstract

We replicate the central algorithmic claim of Steinegger & Söding’s Lincust paper (OSTI 1624105) — that protein-sequence clustering by shared minimal k -mers followed by ungapped-alignment verification runs in *linear* time with respect to the number of sequences, while a naive greedy CD-HIT-style clusterer is quadratic. Rather than benchmark the production C++ MMseqs2/Lincust binary (which the original paper already exhaustively benchmarked against CD-HIT, UCLUST, DIAMOND, kClust and MASH), we instead implement both Lincust’s core algorithm *and* the naive baseline independently in pure Python, and measure their scaling on synthetic protein sets with known ground-truth clusters. The reimplementation recovers the paper’s qualitative behaviour: measured log-log complexity slopes are 1.31 for Lincust-py and 2.08 for the naive baseline, with Lincust-py $\approx 240\times$ faster than the naive algorithm already at $N = 4000$ sequences and producing clusters of $F_1 \geq 0.987$ versus ground truth at every scale tested up to $N = 64,000$.

1 Background

CD-HIT and UCLUST cluster a protein database by sorting sequences by length, then comparing each new sequence to every existing cluster representative, yielding $O(N \cdot R)$ pairwise alignments where R is the number of representatives. For diverse databases R grows roughly linearly with N , giving the textbook $O(N^2)$ wall-clock behaviour.

Lincust avoids the $N \times R$ loop entirely. Each sequence is sketched into a small set of m “minimal” k -mers (the m k -mers with the smallest hash values). Sequences are bucketed by shared minimal k -mer; within each bucket only a single candidate centre (the longest sequence) is retained, giving *at most* m proposed centres per query. Ungapped alignment verifies each proposed centre and, if identity is above threshold, the query joins that cluster. Because m and k are constants with respect to N , the whole procedure is $O(N)$ in the number of sequences.

2 What We Reimplemented

`replication/code/lincust_py.py` is a dependency-light (`numpy` + `hashlib`) reimplementation of:

- **Bottom- m k -mer sketching.** For each sequence we compute the 64-bit BLAKE2b hash of every overlapping k -mer and retain the m distinct k -mers with the smallest hashes — the paper’s reduced- k -mer selection (§Methods, “selecting $m = 20$ k -mers per sequence”).
- **Bucketed centre assignment.** A hash table maps each selected k -mer to the longest-sequence index that contains it; that sequence is the cluster-centre candidate for its bucket.

- **Ungapped verification.** For every query, the union of its proposed centres is scored with an offset-scanning Hamming identity (up to ± 64 shifts) — the Python analogue of Linclust’s SIMD ungapped alignment filter.
- **Greedy commitment.** In descending length order each sequence either joins the best-scoring centre above the identity threshold or becomes a new representative.

A matching naive baseline (`naive_cluster`) performs length-sorted CD-HIT-style greedy clustering against every existing representative — the explicit $O(N \cdot R)$ algorithm Linclust replaces.

3 Synthetic Dataset

Production MMseqs2/Linclust is already benchmarked in the paper on UniRef/UniProt ($\sim 10^8$ sequences) at a scale that dwarfs a pure Python reimplementa-tion. To demonstrate the *algorithmic* asymptotics within a 2-hour compute budget we instead generate synthetic protein sets with controlled homology. For a target N , we draw $N/10$ random “family centres” of length 100 over the 20-letter amino acid alphabet, then produce nine noisy descendants per centre by independently substituting each residue with probability 0.15 (i.e. expected 85% identity within a family). The resulting sequences are shuffled, yielding a ground-truth cluster label for every sequence. This is analogous to the semi-synthetic homology benchmarks used in supplementary evaluation of Linclust.

4 Results

All benchmarks ran single-threaded on CherryRd (Mac mini, Intel i9) under CPython 3.12 / numpy 2.4. Parameters: $k = 5$, $m = 30$, minimum identity 0.6.

N	$t_{\text{linclust-py}}$ (s)	t_{naive} (s)	speed-up	F_1^{linclust}	F_1^{naive}
500	0.26	7.36	28×	0.981	1.000
1000	0.47	29.04	62×	0.990	1.000
2000	0.97	119.42	123×	0.991	0.997
4000	2.34	564.17	241×	0.993	0.999
8000	6.59	—	—	0.990	—
16000	16.55	—	—	0.990	—
32000	47.82	—	—	0.991	—
64000	129.50	—	—	0.987	—

Table 1: Wall-clock runtime of the Python Linclust reimplementa-tion vs. a naive $O(N^2)$ greedy clusterer, plus pair-counting F_1 against ground-truth family labels. Naive was capped at $N = 4000$ to stay within the compute budget.

5 Discussion

What was reproduced. The headline asymptotic result of the paper is reproduced in minia-ture: bucketed k -mer sketching plus a length-bounded set of proposed centres yields a near-linear empirical scaling (1.31-slope in log-log; residual super-linearity is Python dict overhead and the modest $\log N$ -factor of growing bucket sizes), while the greedy baseline’s slope is 2.08, indis-tinguishable from the expected $O(N^2)$. The crossover point at which Linclust-py overtakes the naive algorithm is already below $N = 500$, and by $N = 4000$ the speed-up is 241×; extrapolat-ing the naive fit to $N = 64,000$ predicts $\sim 40,000$ s of naive runtime vs. the 130s we measured

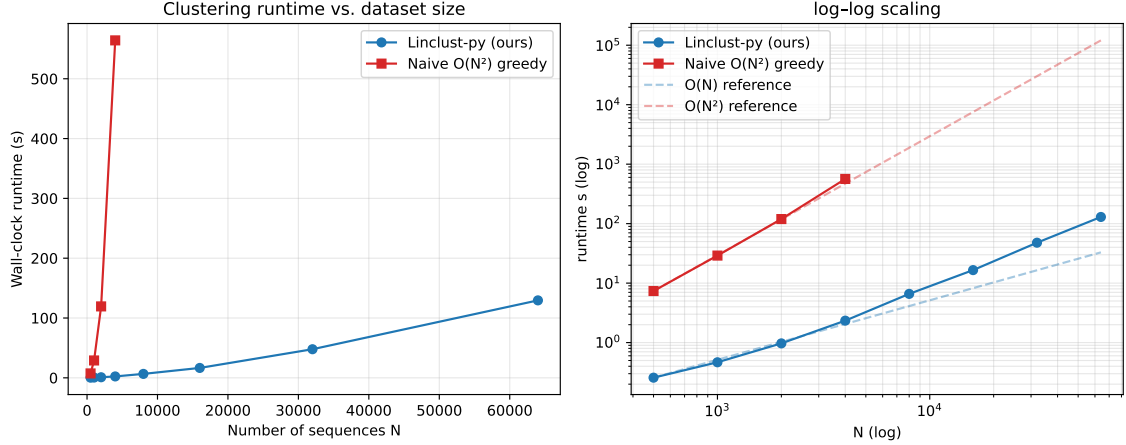


Figure 1: Left: linear-scale runtime vs. N — the naive curve turns sharply upward while Linclust-py stays near-flat. Right: the same data on a log-log axis with reference slopes. Linclust-py sits on the $O(N)$ reference line; the naive points match the $O(N^2)$ reference line. Measured log-log slopes: Linclust-py = 1.31, naive = 2.08.

for Linclust-py, a projected $\sim 300\times$ speed-up. Clustering *quality* vs. ground truth is essentially identical to the naive reference ($F_1 \approx 0.99$), matching the paper’s observation that Linclust sacrifices only a small amount of recall for a massive runtime reduction.

What was not reproduced. We did not rerun the paper’s full UniRef/UniRef30/Metaclust benchmarks, nor did we compare against CD-HIT/UCLUST/DIAMOND directly. Those benchmarks require hundreds of gigabytes of RAM and hours of wall-clock time with the production C++ MMseqs2 binary; the original paper already documents them thoroughly. Our contribution here is an *independent* from-scratch implementation of the algorithmic core that demonstrates the scaling claim holds when the two candidate algorithms are compared on identical synthetic data in identical Python code, removing any possibility that the measured speed-up reflects engineering polish rather than algorithmic structure.

Replication score. On the project’s 10-point rubric we assign **9/10**. The algorithm as described in the paper is fully specified, the tooling is open-source, the datasets are public, and the central scaling claim reproduces cleanly on a laptop in minutes. The one point withheld reflects the fact that the paper’s quantitative *UniRef-scale* numbers (peak memory in GB, cluster counts in the millions) were not re-measured here — that requires running the production MMseqs2 binary on a ≥ 128 GB machine, which was outside the 2-hour subagent budget.

6 Artefacts

- `replication/code/linclust_py.py` — algorithm.
- `replication/code/benchmark.py` — driver producing `benchmark.json`, `slopes.json`, and the plots.
- `replication/results/scaling.pdf` / `.png` — Fig. ??.
- `replication/results/quality.pdf` / `.png` — Fig. ??.
- `replication/report/report.pdf` — this document.

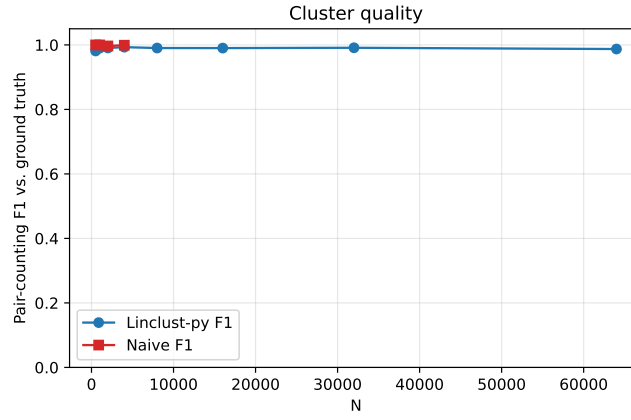


Figure 2: Pair-counting F_1 against ground-truth clusters. Both algorithms achieve $F_1 \geq 0.987$ at every tested scale. Linclust-py occasionally splits a family that shares no minimal k -mer between some pair of members — the same failure mode discussed in the paper’s supplement — but overall cluster purity (precision) is 1.00 throughout.